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Neuro-symbolic artificial intelligence in accelerated design for 4D printing: Status, challenges, and perspectives

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ABSTRACT

4D printing enables the creation of adaptive and reconfigurable devices by combining additive manufacturing with smart materials. This integration introduces challenges in designing printable, responsive materials and structures. Current research focuses on improving the responsiveness and mechanical performance of smart materials, but incremental advances often lack sufficient feedback for achieving specific properties, shapes, and performance targets. Inverse design has emerged as a strategy for determining material compositions and structural configurations to meet desired outputs, but its application remains limited to simple structures. Accelerating material and structural discovery is crucial for advancing 4D printing. Artificial intelligence (AI), especially machine learning (ML), offers promising solutions to address the complexity of 4D printing design. However, conventional AI approaches often lack logical reasoning, explainability, and interpretability. This review paper highlights recent achievements and challenges in 4D printing design and introduces neuro-symbolic AI as a promising approach. By combining ML's learning capabilities with the logical reasoning and semantic understanding of symbolic AI, this approach can enhance the exploration of advanced active materials and structures. The insights provided aim to guide future research toward optimizing 4D printing for broader applications and enhanced performance.

1. Introduction

4D printing is a manufacturing paradigm that combines additive manufacturing (AM) with smart materials (SMs). This technology allows printed objects to change their shape, properties, or functionality over time when exposed to stimuli (e.g., temperature, light, moisture, magnetic/electric field, chemistry, or pH) [159,52]. Through shape changes, the objects can then be programmed to autonomously expand/contract, fold, bend, or morph without the need for mechanical intervention (**Fig. 1**). Since then, the field has seen rapid advancements, driven by incremental innovations in materials science, AM, chemistry, and computational mechanics, which has opened potential applications in various sectors such as biomedical, soft robotics, textiles, automotive, defense, and space [35,36].

At the core of 4D printing lies the integration of stimuli-responsive materials such as shape memory polymers (SMPs), liquid crystal elastomers (LCEs), shape memory alloys (SMAs), electroactive polymers (EAPs), and hydrogels. These shape-changing materials can be processed using various AM techniques, including fused filament fabrication (FFF), direct ink writing (DIW), digital light processing (DLP), and power-bed fusion (PBF), in a way to build sophisticated and functional geometries [38]. For instance, SMPs can be programmed to retain a temporary shape at lower temperatures and then recover their original shape when heated above a certain threshold [86]. Similarly, hydrogels can swell or shrink in response to variations in humidity, temperature, ion concentration, or pH, making them particularly suitable for biomedical applications [51,1,143,191,185,42]. Each active material exhibits

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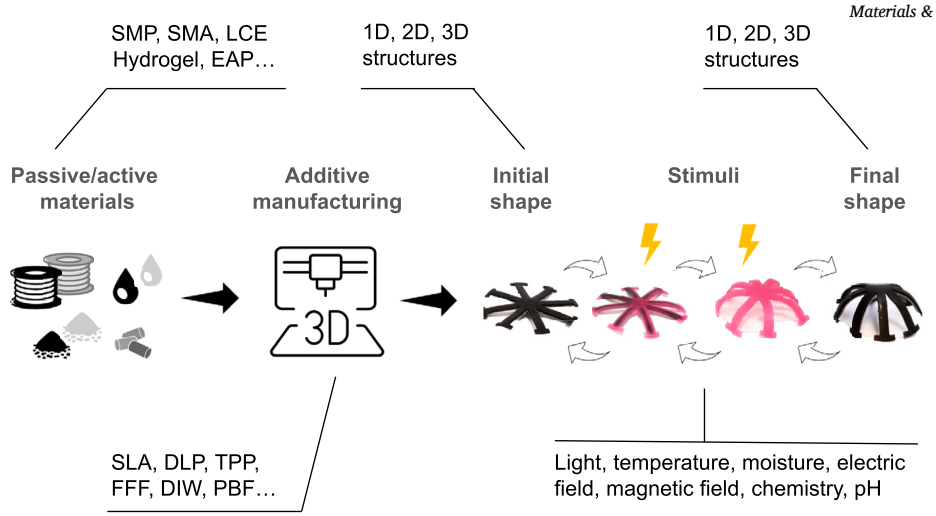

Fig. 1. 4D printing paradigm illustrated through a shape- and property-changing structure.

particular advantages depending on the intended use and the available stimuli [15].

Current research in 4D printing focuses on enhancing the printability, stimuli sensitivity, functional fatigue resistance, and mechanical performance of SMs [34]. However, despite these advancements, significant challenges remain in predicting and controlling the behavior of stimuli-responsive materials, especially when targeting complex properties and functionalities [112,15]. Achieving precise control over material transformations is essential to ensure that 4D-printed structures perform as intended in real-world applications. Another critical challenge lies in the development of multi-material structures, where materials with diverse mechanical, thermal, and chemical properties must bond effectively and work cohesively under stimulation [175]. To address these challenges, it is necessary to leverage computational intelligence that account for the dynamic, time-dependent interactions between active and passive materials. Indeed, the development of computational intelligence methods can accelerate the exploration and discovery at both the material and structure design levels, while improving their printability and functional performance over time.

Advanced methodologies such as inverse design have emerged, allowing mechanical engineers and designers to work backward from target material properties, shapes, or structural behaviors in order to determine the optimal design architecture and material composition [100,148,154]. This enables the creation of tailored materials and structures that can meet specific performance criteria. Machine learning (ML) has become indispensable in inverse design as it offers the capacity to model complex datasets and predict material behavior, thus accelerating the discovery and optimization cycles [19]. By transforming raw data into actionable insights, ML aids in decision-making for material selection and design. However, traditional ML techniques often required manual feature extraction, which is both complicated and time-consuming. As such, the rise of deep learning (DL) has automated feature extraction using neural networks (NNs), leading to major breakthroughs in various fields such as computer vision, natural language processing (NLP), and autonomous driving, where large datasets need to be processed efficiently [187].

Deep Learning (DL) fundamentally relies on NNs, which consist of interconnected layers of artificial neurons that process and transmit information. As inputs are propagated through the layers, the network learns complex patterns and improves its predictive accuracy over time. However, while NNs are highly effective at pattern recognition, they encounter challenges in tasks that require logical reasoning, explainability, and interpretability, which are essential for high-stakes design and engineering applications [109]. To address these limitations, symbolic AI presents a complementary approach via a structured, rule-based reasoning process. By combining these two AI methods, neuro-symbolic AI (NSAI) has emerged as a promising paradigm which considers the dual-process theories of human cognition [139,48,76]. In these theories, the fast and intuitive processing is represented by sub-symbolic NNs, while deliberate, logical thinking aligns with symbolic reasoning systems. NSAI integrates these complementary modes of processing, which enables systems to learn from experience while reasoning based on structured knowledge, thus bridging the gap between intuitive learning and explicit logical reasoning [13,49,83,101,163,134,184]. By combining the learning capabilities of NNs with the logical structure of symbolic reasoning, NSAI creates a hybrid architecture that not only improves decision-making efficiency but also enhances the explainability of results. In the context of design for 4D printing, NSAI allows for the generation of explainable recommendations by leveraging symbolic reasoning to break down complex material interactions into understandable, rule-based decisions. This makes it easier to trace and understand the reasoning behind specific design decisions related to active material composition, formulation, selection, and distribution in the 2D/3D design space.

Therefore, this paper aims to explore new opportunities in design for 4D printing and guide future research focused on enhancing the properties of active materials and performance of structures to unlock real-world applications. Section 2 reminds the main challenges in design for 4D printing in terms of printable active materials and inverse design methods. Section 3 studies the potential of NSAI as a transformative tool for addressing these challenges. It highlights the benefits of combining ML with symbolic reasoning for enhanced explainability and interpretability. Then, Section 4 outlines critical future research directions needed to accelerate the development of next-generation SMs and transformable structures as well as the discovery of latent applications.

2. Status and challenges in 4D printing design

2.1. Printable and stimuli-responsive materials

SMs play a pivotal role in advancing 4D printing as they allow structures to dynamically respond to stimuli. Despite their potential, these materials still face limitations that hinder their widespread application [38]. For instance, SMPs are valued for their lightweight nature, cost-effectiveness, and shape recovery capabilities. However, the mechanical training requirements and lack of reversibility make them less suitable for applications requiring repeatable actuation [3,128,47,182,132,144]. LCEs, known for their large deformation capabilities and rapid, reversible actuation, are well-suited for dynamic and repeatable applications. Nonetheless, they encounter challenges with high-resolution

printing and exhibit functional fatigue over multiple actuation cycles [130,183,162]. Hydrogels, widely used in biomedical contexts, can effectively interact with biological fluids, yet they suffer from low mechanical strength, long response, and recovery times although reversible. Their tendency to dry out in non-aqueous environments further limits their reliability in diverse conditions [133,26,29]. Last, SMAs, such as nickel-titanium alloys, provide robust, reversible actuation upon heating. However, their brittleness, long response times, and nickel-related toxicity concerns complicate their use, especially for biomedical applications [21,102,4].

These limitations require further effort in material composition, formulation, processing, and the creation of composites that blend the strengths of different SMs to expand their capabilities and applications across diverse fields. In addition to this, researchers in materials science are increasingly focused on discovering new SM compositions that enhance their printability, sensitivity to stimuli, functional durability, and mechanical performance. Advancing these materials is crucial for improving the adaptability and efficiency of 4D-printed structures, enabling faster and more reliable responses to environmental changes. A key challenge is overcoming functional fatigue, which currently restricts the reliability of many SMs in applications requiring repeated or sustained use [92,80,12]. Developing the next-generation of printable SMs is then essential for ensuring the adoption of 4D printing in the industry. This effort involves engineering materials that exhibit faster actuation, greater mechanical strength, and enhanced responsiveness while also addressing critical concerns like biocompatibility and reducing toxicity risks, especially in medical and sensitive environments. As researchers push the boundaries of materials science and AM, the potential of SMs in fields like healthcare, space, automotive, and soft robotics continues to grow, opening new doors for innovative and adaptive solutions.

However, the development of SMs is intrinsically tied to their responsiveness to stimuli. The stimuli sensitivity plays a critical role in optimizing energy efficiency and ensuring precise control over material behavior. Each stimulus interacts with SMs in unique ways, enabling transformations in shape, function, or physical properties, and making them versatile to a variety of applications. For example, temperature can be used to precisely control deformations in materials such as SMPs, LCEs, hydrogels, and SMAs, allowing them to shape change in a predictable manner. Light, particularly ultraviolet (UV) and near-infrared (NIR), can rapidly trigger localized changes in material properties, making them ideal for high-resolution applications. Similarly, water and pH levels influence hydrogels that can expand or contract, which is especially useful in the biomedical field for drug delivery or tissue engineering applications. Electric fields, on the other hand, provide fast and precise control over material transformations, making them suitable for the development of artificial muscles. Meanwhile, magnetic fields offer the advantage of non-invasive, remote control, which is particularly effective in soft robotics and medical devices where safety and precision are paramount [67,176,69].

Beyond formulating and printing single active materials, researchers are exploring the integration of multiple materials within a single design to leverage the complementary properties of the involved materials [103]. These advancements highlight the growing need to optimize material distribution within structures in order to achieve sophisticated shapes and actuation performance. This involves more than simply combining active and passive materials, but also requires a deep understanding of how different materials interact via stimuli and how their arrangement influences the overall behavior of the structure. By carefully tuning material placement, it is possible to fabricate objects that not only respond dynamically to stimuli but also meet specific mechanical and transformation requirements, such as strength, flexibility, and reversibility [132]. To achieve targeted shape changes, advanced computational design approaches are essential to predict the right material at the right place in the 2D/3D structure. This strategy opens up exciting possibilities for the discovery of new digital material distributions, which could expand the capabilities of SMs.

2.2. AI-enabled inverse design

Through inverse design, material composition and distribution, printing parameters, and desired properties can be systematically determined to achieve precise target shapes and performance. In the field of materials science through materials by design, significant progress has been made in utilizing ML for predicting and optimizing the mechanical properties of composite and heterogeneous materials. For example, convolutional neural networks (CNNs) have been used to predict the effective modulus of these composites, allowing for rapid design iterations [22,93]. Regarding strength and toughness, Gu et al. [55] and Yu et al. [181] developed ML classification and regression models that optimize the mechanical performance of composites under shearing. Hierarchical design approaches have been developed to consider microstructures as building blocks (pixels) to predict mechanical properties [56].

Hybrid methods that integrate ML with evolutionary algorithm (EA) have proven successful in optimizing shape-change responses in 4D-printed active composite structures, offering faster and more efficient inverse design solutions [155]. For instance, finite element (FE) simulations combined with EA have been used to explore the vast design space of active composite beams with voxel-level material distributions, leading to significant advancements in shape optimization [57,147,148]. While traditional EA approaches are computationally expensive, ML models, such as ResNet and inverse NN, have been integrated to drastically reduce computational costs, enabling quicker and more accurate predictions of shape-change responses [154,104,153] (**Fig. 2a,b**). These models have demonstrated remarkable success in handling complex, high-dimensional inverse design problems, such as buckling mesosurfaces and inflatable composite membranes [46,28]. Athinarayanarao et al. [9] extended the use of voxel-based approaches by incorporating void as a third material in EA to optimize the use of active material in 3D active composite structures (**Fig. 2e**). As for stress and strain fields prediction, conditional generative adversarial networks (GANs) have been employed in materials with pixel-level material distribution. For example, Yang et al. [178] demonstrated how conditional GANs could predict stress and strain fields in hierarchical composites, handling varying component shapes, boundary conditions, and material hierarchies. These models have significantly improved the generalization and robustness of designs under various conditions, so as to enable more efficient and reliable optimization of composite structures.

Furthermore, physics-informed neural networks (PINNs), introduced by Raissi et al. [127], have been applied to inverse problems like material characterization, where they enable the identification of heterogeneous material properties based on observed deformation data. This innovative approach integrates physical laws directly into the ML models in order to improve precision in material property identification. Such integration not only reduces the need for extensive labeled data but also enhances the overall efficiency and accuracy of predictions. Mechanical metamaterials have seen the application of deep generative models such as GANs and variational autoencoders (VAEs) to design architectures with tailored anisotropic elastic properties. These models allow for the rapid exploration of new microstructural designs with enhanced stress-strain responses, then facilitating the discovery of novel metamaterials with target mechanical properties [108,25,24,23,171,172,170]. Despite these research efforts, the resulting material distributions may introduce printability issues, especially when dissimilar materials are employed. To overcome this barrier, Benyahia et al. [14] developed a computational design approach that converts any voxel-based distribution achieving a desired shape change into interlocking blocks suitable for manual or robotic assembly (**Fig. 2d**). Adding to these advances, Jin et al. [74] proposed a hybrid optimization framework for ultrafast inverse design of 4D-printed active composite structures. Sun et al. [152] used recurrent neural networks (RNNs) for the design of active composite structures, enabling the precise prediction of material distribution that react dynamically to environmental stimuli (**Fig. 2c**).

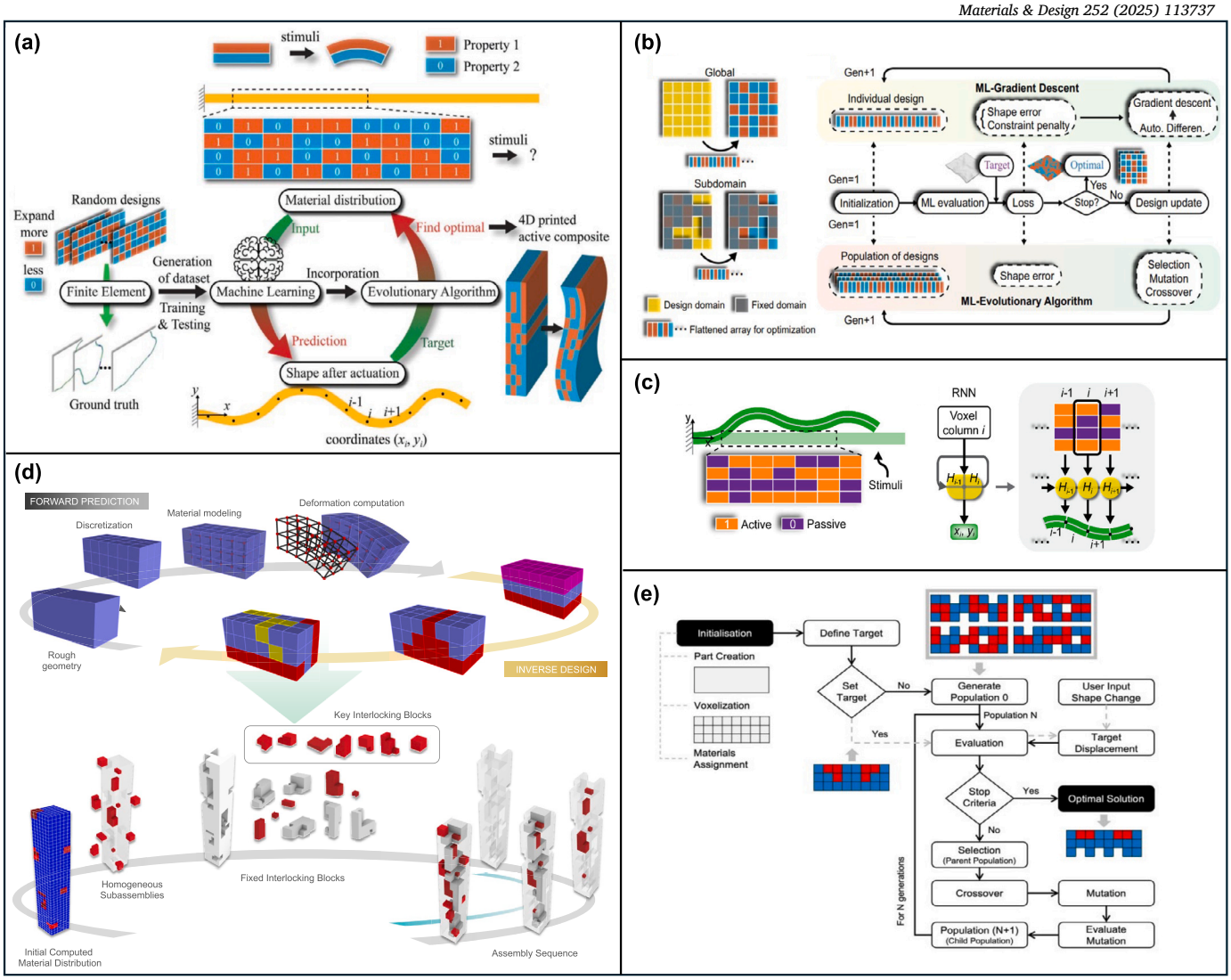

Fig. 2. Inverse design and optimization architectures of 4D-printed multi-material structures. (a) Design of 4D-printed active composite beam using ML-based prediction and EA-based optimization [154] (Reproduced with permission from Wiley). (b) Inverse design strategies based on ML and EA [153]. (c) Active composite determination using RNNs [152] (Reproduced with permission from Elsevier). (d) Computational design for building multi-material structures using interlocking blocks and voxel-based modeling [14] (Reproduced with permission from Elsevier). (e) EA-driven topology optimization of active composite structures [9].

Recent efforts have also explored ML applications in soft robotics, where sustainable robotic joints with variable stiffness have been developed using reinforcement learning to predict and optimize the behavior of 4D-printed structures. This enabled the precise actuation and control of soft robotic systems [111]. ML was employed in the design of dielectric elastomer to fabricate actuators that can respond to electric field in order to enhance the potential for human-machine interaction [79]. Soft pneumatic actuators demonstrated that ML can significantly enhance the precision of shape changes through time-efficient predictions compared with numerical simulations [189]. Additionally, AI-based control systems have been integrated with 4D printing processes to address the nonlinear behavior of active materials, improving the adaptability and precision of actuators and paving the way for more autonomous and intelligent structures [15].

These initiatives highlight the growing importance of ML in advancing 4D printing by accelerating material discovery, improving design and optimization processes, and enhancing the performance of 4D-printed structures across various applications. However, to fully harness the potential of ML, especially when working with small datasets or when detailed explanations are required in quantitative research, integrating symbolic AI becomes crucial. This is particularly relevant for the emerging field of 4D printing. Symbolic AI complements data-driven approaches by structuring background knowledge and providing more transparent and interpretable results. This integration ensures that ML-driven discoveries in 4D printing are not only accurate but also easily validated, making them more applicable to real-world scenarios.

Complementary research efforts have been made on symbolic AI for 4D printing. Dimassi et al. [41] have developed an ontology for structuring and formalizing knowledge related to stimuli, SMs, product design, and AM. By leveraging this domain ontology, researchers can ensure that the captured knowledge is not only accessible but also interoperable with other domains, as advocated by the Industry Ontology Foundry (IOF) initiative [44]. Ontologies also serve as a foundation for the development of reasoning procedures, offering qualitative decision support in material and process selection, design recommendations, and the alignment of 4D printing practices with emerging standards. By enabling real-time updates and reasoning over structured knowledge, ontologies can bridge the gap between symbolic AI and neural AI approaches, providing a more holistic and context-aware design strategy [85] [37]. Effectively integrating this structured knowledge into a broader design and material informatics frameworks will be key to the future of 4D printing.

Table 1
Comparison between symbolic AI and neural AI.

Aspect	Symbolic AI	Neural AI
Generalization	Strong at handling abstract concepts and generalizing beyond training data (out-of-distribution reasoning)	Weak in generalizing beyond training data, primarily excels within the dataset it has seen (overfitting potential)
Data handling	Struggles with unstructured data (e.g., images, raw text)	Excels at handling unstructured data (e.g., visual, auditory, textual data)
Learning speed	Slow learning, requires pre-defined rules	Fast learning from large datasets, can learn patterns without explicit rules
Interpretability	Highly interpretable (decisions can be traced to explicit rules and logic)	Lacks interpretability (black-box models, hard to understand the decision-making process)
Robustness	Weak robustness, brittle in noisy environments or with incomplete data	Highly robust, adaptable to noisy and varied data environments
Scalability and optimization	Limited scalability, struggles with large-scale, complex datasets	Highly scalable, efficient in large-scale data processing and optimization
Reasoning and explainability	Strong reasoning capabilities, excels in logical reasoning and providing explanations	Weak reasoning and explainability, struggles to provide transparent logic behind decisions
Data requirements	Requires less data, relies more on rules and knowledge bases	Requires large labeled datasets for training, data-intensive
Uncertainty and inference	Struggles with uncertainty, less flexible in probabilistic inference	Strong at learning and inference under uncertainty, better at probabilistic reasoning

As the complexity of 4D printing continues to increase, the need for explainable and traceable recommendations becomes ever more critical. Transparent decision-making tools are vital, especially in fields such as materials science, engineering, design, and manufacturing. By combining the data-driven capabilities of ML with the logical reasoning of symbolic AI, it is possible to create more transparent and scientifically grounded design insights. This fusion enables more efficient, reliable, and scalable 4D printing processes, fostering innovations in multiple industrial sectors.

3. The rise of neuro-symbolic AI

NNs have shown remarkable proficiency in processing unstructured data such as images, audio, and text. Their ability to learn complex patterns and representations from vast datasets has led to significant breakthroughs in various fields, including computer vision, speech recognition, and natural language processing (NLP) [78,166]. One of the key strengths of NNs lies in their adaptability and capacity to learn from raw data without requiring predefined rules or expert knowledge. This makes them highly scalable and effective in applications that involve large amounts of unstructured data. However, despite these strengths, NNs have several notable weaknesses. One of the most significant issues is their lack of transparency. Neural models are difficult to interpret, making it challenging to understand how they arrive at specific decisions or predictions. This opacity poses problems for critical applications where explainability is crucial, such as healthcare, finance, legal systems, or engineering. Additionally, NNs are highly data-hungry, requiring vast amounts of labeled training data to perform effectively. This reliance on large datasets makes them inefficient in scenarios where data is limited or costly to acquire. Furthermore, neural models struggle with reasoning and generalizing beyond their training data, which leads to poor performance in tasks requiring logical inference or commonsense reasoning. Specifically, tasks such as understanding causal relationships, solving problems involving sequential logic, and making decisions that rely on external world knowledge (**Table 1**).

Symbolic AI excels in areas where NNs fall short. Symbolic systems are based on explicit rules and structured representations, which enable them to perform complex reasoning tasks, such as mathematical proofs, planning, and expert systems. The transparency of symbolic AI is one of its greatest strengths. Since symbolic methods rely on predefined rules and logical structures, their decision-making processes are easily interpretable and explainable. However, symbolic AI has its own limitations. One of the major drawbacks of symbolic approaches is their rigidity and difficulty in adapting to new scenarios. Symbolic systems require manually crafted rules and structured input data, making them difficult to scale in real-world environments where data is often noisy, incomplete, or unstructured. They are also vulnerable to combinatorial explosions when dealing with large datasets or complex reasoning tasks, which significantly hampers their performance at scale. Moreover, symbolic systems are not well-suited for tasks involving perception, such as image or speech recognition, as they cannot autonomously extract knowledge from unprocessed data (**Table 1**).

3.1. Advantages of NSAI

While traditional NNs are great at recognizing patterns in data, they often struggle when faced with situations they have never seen before. On the other hand, symbolic reasoning provides a logical framework for decision-making but is constrained in its ability to acquire knowledge from new data and adapt dynamically. By bringing these two approaches together, NSAI manages to fill the gaps and create a more adaptable, interpretable, and efficient AI system (**Fig. 3**).

One of the standout features of NSAI is its ability to generalize beyond its training data. Traditional AI models often falter when encountering something new, but NSAI, with its blend of learning and logic, can handle these situations better. This quality is essential in real-world applications such as autonomous vehicles or healthcare, where systems must reliably function even in unpredictable environments. In addition, NSAI can offer better interpretability of the decision-making process in designing for 4D printing by providing insights into how and why specific design decisions are made, making this approach both more trustworthy and valuable. Another significant benefit of NSAI is its reduced need for large datasets. Traditional AI-based systems often require massive amounts of data to perform well, which can be costly in terms of both time and resources. However, NSAI can achieve high performance with far less data, thanks to its ability to reason symbolically. This makes it a more sustainable and practical option, especially for smaller organizations or emerging research field working with limited resources. In addition to this data efficiency, NSAI models also show remarkable transferability, meaning they can take what they have learned in one area and apply it to another with minimal retraining. This is incredibly useful in situations where there is not much data available for a new task.

What really sets NSAI apart is its enhanced reasoning capabilities. While NNs are proficient in detecting correlations and underlying structures within datasets, they often fall short in tasks requiring logical reasoning or problem-solving. NSAI bridges this gap by handling more complex challenges, whether it is diagnosing a medical condition, analyzing legal cases, or making scientific discoveries. Additionally, NSAI is highly scalable and adaptable, making it capable of tackling increasingly

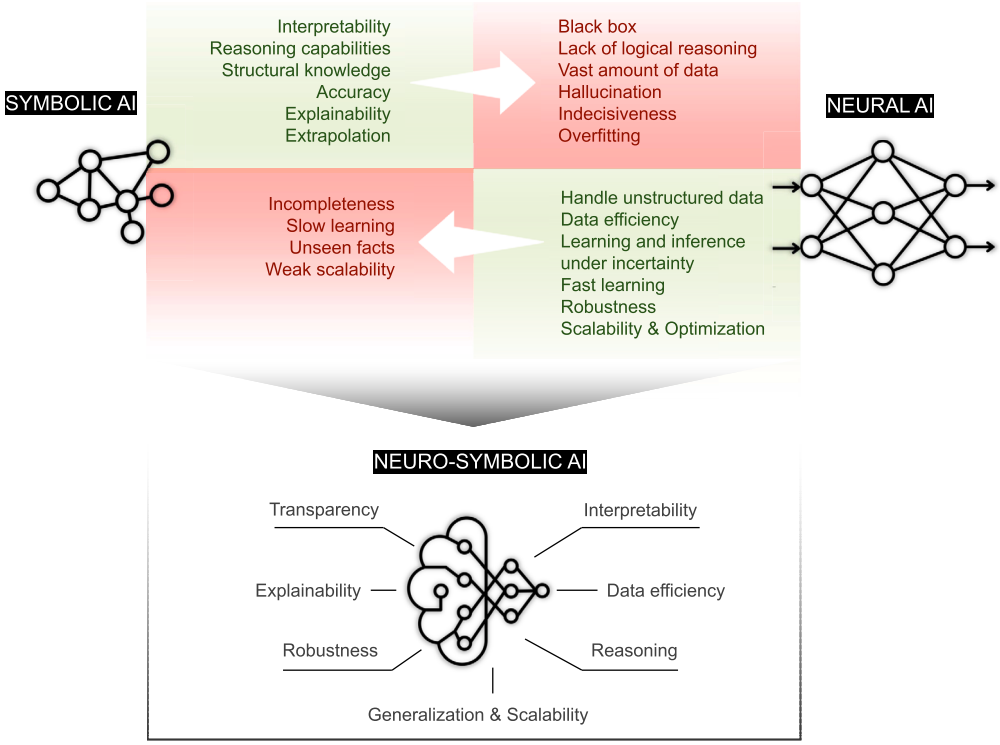

Fig. 3. Neuro-symbolic AI as a combination of symbolic AI and neural AI strengths (illustrated in green parts).

complex problems across various fields. Its ability to combine unstructured data processing with structured logic means it can continue to grow and evolve as AI technologies advance, positioning NSAI as a key player in the future of AI-based systems.

3.2. Taxonomy of NSAI

The taxonomy of NSAI presents the main elements that combine NNs and symbolic reasoning to build a hybrid approach. It shows how neural components are designed to learn from large datasets, while symbolic components offer logical frameworks that support decision-making and explainability. By categorizing this AI paradigm, researchers and practitioners can understand how these two pillar systems work together in a way to address real-world applications.

3.2.1. Neural component

The neural component serves as the foundation for data-driven learning that is crucial for tasks like vision, language processing, and reasoning. Numerous neural architectures have been investigated and developed to advance the capabilities of NSAI, as illustrated in Fig. 4. NNs, such as CNNs, are important in handling grid-like data, particularly images, by learning local patterns through convolutional layers. These CNNs have been fundamental in revolutionizing fields like computer vision, enabling tasks such as image recognition and object detection with their hierarchical feature extraction.

Another important architecture is the multilayer perceptron (MLP), one of the earliest and simplest NNs. MLPs are built with fully connected layers and are effective for classification and regression tasks, especially when the relationships between input and output are straightforward. Although more complex models have emerged, MLPs remain valuable in simpler tasks due to their efficiency and ease of use. For graph-structured data, graph neural networks (GNNs) excel by processing relationships between nodes, allowing for advanced reasoning in areas such as social networks, molecular chemistry, and recommendation systems. GNNs leverage node connections to perform neighborhood aggregation in order to help the network learning meaningful patterns from graph-based inputs. In tasks where sequences are involved, Sequence-to-Sequence (Seq2Seq) models have made significant strides [180]. These models manage both input and output sequences using an encoder-decoder structure in tasks such as language translation and time-series analysis. Many improvements have been made to enhance their accuracy such as attention mechanisms [166]. Transformers, a category of seq2seq models, have revolutionized NLP by using a self-attention mechanism that allows for parallel data processing. Based on the transformer architecture, models, such as BERT [40], RoBERTa [98], and T5 [126], have achieved promising results in question answering, text generation, and machine translation. Larger versions of these models, such as ChatGPT [106] and PaLM [94], have expanded the possibilities of AI, therefore showing potential for applications like education and personalized recommendations, and hinting at their future role in artificial general intelligence.

More recently, while large language models (LLMs) have demonstrated impressive performance in various tasks. Their ability to generate reliable factual knowledge remains a concern. They tend to memorize facts from the training data but struggle to recall them accurately, which often lead to hallucinations with incorrect statements [151,87,169]. For example, an LLM might claim that “Einstein discovered gravity in 1687”, which is not true as Isaac Newton formulated gravitational theory. This undermines the trustworthiness of LLMs. Moreover, seen as black box models, LLMs are difficult to interpret due to their implicit representation of knowledge within their parameters [91]. This lack of interpretability, combined with their probabilistic reasoning processes, makes their predictions and decisions difficult to validate or explain [190]. Even when LLMs provide explanations using techniques such as chain-of-thought reasoning [71], these explanations can still suffer from hallucination issues [167]. This severely limits the application of LLMs in high-stakes engineering applications like 4D printing, where inaccurate reasoning could have significant consequences on the multiple inherent aspects of this technology. Additionally, training LLMs on broad datasets may face difficulties in effectively generalizing to specialized fields or integrating new knowledge, as they often lack domain-specific information [106]. LLMs are based on the

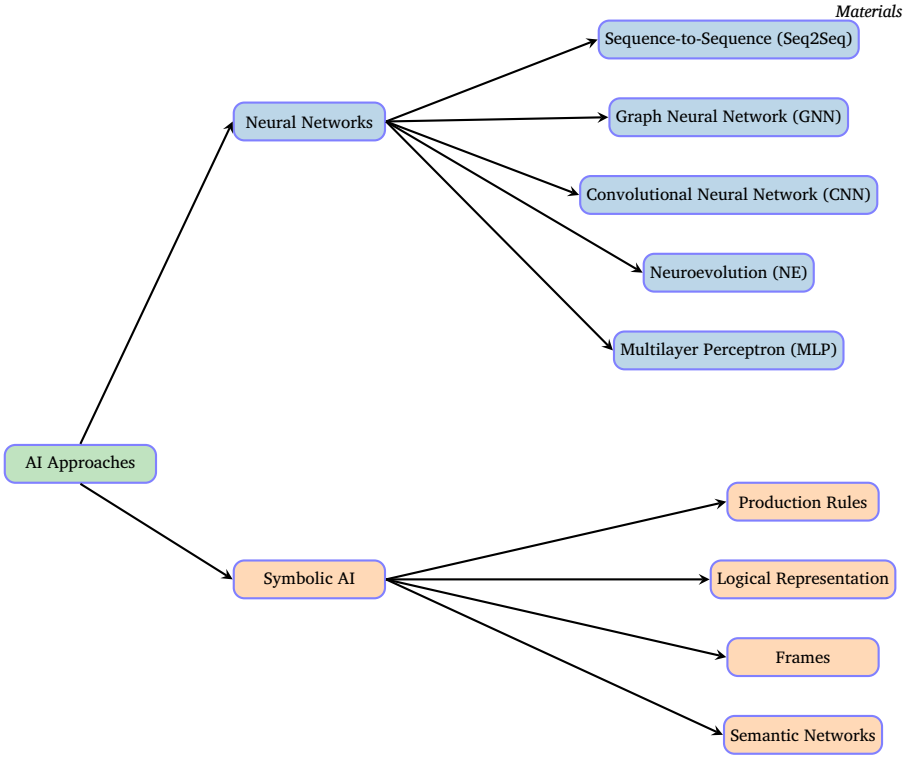

Fig. 4. Existing neural and symbolic AI architectures.

transformer architecture, which utilizes an attention mechanism. The latter enables the model to focus on the most important parts of the text during processing. Transformers consist of two main components: the encoder and the decoder. In the context of language models, the decoder often plays a central role.

In an encoder, the input text – such as a question or sentence – is first tokenized. The tokenization breaks the text into smaller units called "tokens", which can be words or sub-words. Each token is then converted into a numerical representation, called an embedding, using an embedding model. These embeddings are vectors in a multidimensional space, where tokens with similar meanings are positioned close to each other, capturing the semantic relationships between them. For instance, the prompt "What are the key properties of hydrogel?" can be converted into a sequence of numerical tokens. These tokens are then represented in a vector space where words like "hydrogel" and "swelling" are placed near each other due to their contextual relationship. Once the embeddings are generated, they are fed into the pre-trained transformer model. Using its attention mechanism, the model focuses on the relationships between different tokens while considering the overall context of the sentence. This allows it to understand not only the individual meanings of the words but also their interactions within the sentence. One of the strengths of the transformer architecture is its ability to capture long-range dependencies in text. This means the model can grasp how words that are far apart in a sentence influence each other, contributing to a deeper understanding of the sentence's meaning. In the decoding phase, the decoder generates the output text one token at a time. Based on the context provided by the encoder and previous predictions, the decoder predicts the next word in the sequence. For instance, after "Hydrogel exhibits", the model might predict the word "swelling", demonstrating its understanding of the contextual relationship between "Hydrogel" and "swelling" (see Fig. 5).

Recent years have witnessed significant advancements in the field of multimodal large language models (MLLMs), driven by the complementary strengths of LLMs and large vision models (LVMs). MLLMs bridge the gap between text-based reasoning and visual understanding, enabling models to process, reason, and generate content across multiple media, including text, images, videos, and audio [173,97]. These models leverage the vast pre-training of LLMs, incorporating visual information to enhance their reasoning capabilities, thus overcoming the inherent limitations of conventional LLM models. MLLMs exhibit two distinct traits that set them apart from traditional models: (i) they are built on LLMs with billions of parameters, which offer unprecedented generalization and scalability [186], and (ii) they employ novel training paradigms such as multimodal instruction tuning [173,97]. This combination empowers MLLMs with unique capabilities, such as generating website code from images [186], interpreting complex memes [177], and performing optical character recognition (OCR)-free mathematical reasoning [43]. The launch of GPT-4 marked a pivotal moment, which showcased the potential of MLLMs and increased research and development efforts from both academia and industry [2].

The progress of MLLMs has been rapid and multifaceted towards applications across various domains. These include enhanced support for fine-grained control in user prompts [27], better handling of diverse input and output modalities [59,114], and extension to multilingual contexts [65,10]. Additionally, MLLMs are being adapted for specific applications, such as medical image analysis [88,115] and document parsing [99,73]. As research in this field continues to evolve, MLLMs are poised to play a crucial role in the future of AI, particularly in scenarios requiring complex multimodal reasoning and interaction. Building upon MLLMs, the integration of retrieval-augmented generation (RAG) has emerged as a crucial approach to address several inherent limitations of LLMs, such as hallucinations and knowledge obsolescence. This particularly relevant in the rapidly evolving field of 4D printing, where the volume of new research and published knowledge is increasing at an unprecedented pace. RAG enhances LLMs by incorporating relevant external knowledge, retrieved based on semantic similarity, into the generation process. This significantly improves the factual accuracy and reliability of the outputs [70], especially in knowledge-intensive tasks like designing for 4D printing. RAG operates by retrieving pertinent sections of documents from an external knowledge base and incorporating them into the LLM's response generation. This retrieval process mitigates the risk of generating factually incorrect or outdated content [68]. The development of RAG has led to three main paradigms, namely naive RAG, advanced RAG, and modular RAG, each offering progres-

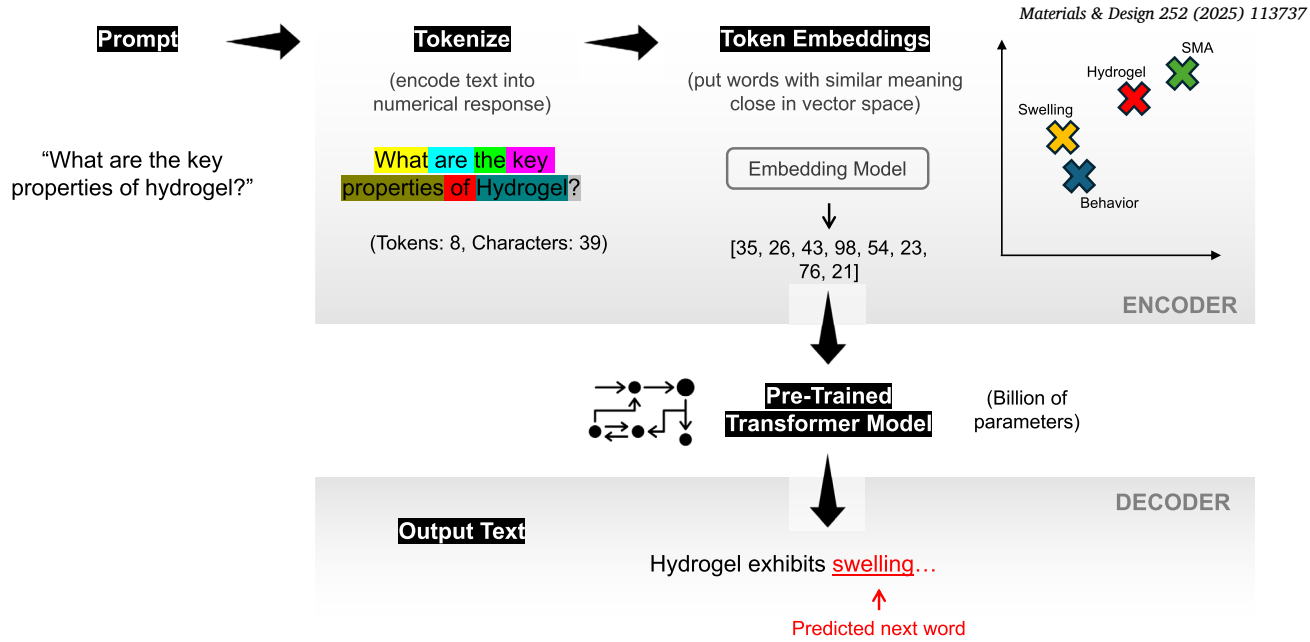

Fig. 5. Process flow of a pre-trained transformer model for text generation in 4D printing.

sive improvements in retrieval and generation capabilities. Naive RAG relies on a basic retrieve-read framework, while advanced RAG incorporates sophisticated pre- and post-retrieval strategies to enhance context relevance. Modular RAG, the most flexible approach, introduces additional specific components that further refine retrieval accuracy and content generation [73]. The integration of RAG into MLLMs addresses the "blindness" of LLMs to non-textual data by incorporating multimodal retrieval capabilities, thus enabling more informed and contextually accurate responses. This synergy between MLLMs and RAG opens up new avenues for high-stakes applications, like those involved in design and 4D printing, where accuracy and factual correctness are paramount [5].

Moreover, neuroevolution is another neural component that represents a pivotal approach in optimizing NNs by leveraging EAs rather than traditional gradient-based methods like backpropagation. This approach relies on genetic algorithms to evolve the topologies, weights, or ensembles of NNs across multiple generations. Neuroevolution is particularly effective in dynamic environments, such as those encountered in reinforcement learning, where predefined architectures may struggle to adapt effectively. By evolving neural structures over time, neuroevolutionary techniques can discover and refine architectures better suited to specific tasks, making them especially valuable in fields like robotics and autonomous systems such as those enabled by 4D printing [58].

3.2.2. Symbolic component

In NSAI, the symbolic component is critical for reasoning and decision-making. It complements NNs by structuring knowledge in a way that allows logical inference and interpretability. This symbolic reasoning can be categorized into four key knowledge representation frameworks: production rules, logical representation, frames, and semantic networks. Production rules are fundamental in rule-based systems, where the knowledge is represented as "If-Then" statements that guide decision-making. Each rule consists of an antecedent (the conditions) and a consequent (the actions to be taken if the conditions are met) [17]. In addition, logical representation provides a framework for reasoning using formal logic systems. It involves defining syntax (symbols and sentence structure), semantics (meaning and truth values), and an inference procedure (rules for deriving new truths). In 4D printing, logical representations can be used to model complex relationships between stimuli and material responses. For instance, we can suppose the following predicates in Table 2.

Table 2
Predicates used to describe logical rules in 4D printing.

Predicate	Description
$A(x)$	"x is an Active Material"
$P(x)$	"x is a Passive Material"
$S(s)$	"s is a Stimulus"
$App(s,x)$	"Stimulus s is applied to Material x"
$I(x,y)$	"Material x interacts with Material y"
$D(x)$	"Material x deforms"
$Time(t)$	"Time t"
$C(x)$	"Material x contracts"
$E(x)$	"Material x expands"
$FinalState(x,t)$	"Material x reaches its final state at time t"

From this predicate logic, several rules can be defined to describe material response to stimuli (Equation (1)), active and passive material interaction (Equation (2)), and time-dependent behavior (Equation (3)) for instance.

$$\forall x, \exists s \; A(x) \wedge S(s) \rightarrow \; C(x) \tag{1}$$

$$\forall x, y, \exists s \; A(x) \wedge P(y) \wedge I(x,y) \wedge S(s) \wedge App(s,x) \wedge App(s,y) \rightarrow D(x) \wedge D(y) \tag{2}$$

$$\forall x, \exists s, \exists t'' > t' > t \;\; A(x) \wedge S(s) \wedge T(t) \rightarrow C(x) \vee E(x) \wedge T(t') \rightarrow FinalState(x) \wedge T(t'') \tag{3}$$

Through these representations, Rule 1 describes that for an active material x under the effect of a stimulus (temperature for instance), the material will contract over time t. Rule 2 makes explicit that if the active material x interacts physically with passive material y under a stimulus, a deformation of both materials occurs. As for Rule 3, it is stated that the active material x may exhibit a time-dependent transformation (contraction or expansion) to reach its final state after the stimulus is applied. These simple rules are capable of capturing various scenarios involving active/passive materials, stimuli, and their interaction over time. They illustrate the flexibility of logical representation to model existing behaviors in 4D printing [45,17].

In addition, frames are objects that encapsulate entities, along with their properties and methods [17]. One type of frame is an ontology, a term that is used in different contexts with various meanings. In the realm of computer science, an ontology refers to a structured represen-

tation of concepts within a specific domain, along with the relationships that connect those concepts [54]. As such, an ontology can be considered as a formal specification of a share conceptualization [6]. The notion of shared conceptualization underscores the importance of consensus among stakeholders, ensuring that the ontology reflects a collective understanding rather than an individual perspective. In practical terms, computational ontologies serve as a means to formally model the structure of systems, encompassing relevant entities and their interrelationships. These models enable various applications, from knowledge representation to automated reasoning, by providing a consistent and machine-interpretable framework for understanding complex domains. Ontologies are often written using the Web Ontology Language (OWL) [164], a highly expressive standard of the Semantic Web that enables detailed descriptions of data structures and advanced reasoning capabilities. Furthermore, standards such as RDF [123], RDFS [168], and SKOS [110] serve as foundational components of the Semantic Web, facilitating data consistency and enhancing interoperability across diverse graphs.

As such, ontologies are not static artifacts but are dynamic constructs that evolve alongside scientific knowledge and the needs of their users. These representations are designed to be both accurate and computable, enabling a wide range of applications from data integration to advanced reasoning. However, the construction and maintenance of ontologies are resource-intensive tasks that require substantial collaborative efforts from domain experts, curators, and ontology specialists. These efforts are essential to ensure that ontologies remain active and reflective of the latest scientific achievements.

In life sciences, for example, ontologies such as the Gene Ontology (GO) [158], Mondo [165], Uberon [117], and FoodON [53] are critical for tasks like gene function annotation, disease classification, and food dataset curation. These ontologies also form the backbone of major data generation projects, including The Encyclopedia of DNA Elements (ENCODE) [33] and the Human Cell Atlas [122]. In 4D printing, ontologies can be incorporated into the frame structure to organize knowledge about SMs, AM techniques, and stimuli. For instance, the HERMES ontology has been developed to formalize the 4D printing knowledge involved in materials science, AM, design, mechanics, and physics [41]. **Fig. 6** presents a portion of the 4D printing ontology, emphasizing the design and engineering view as well as the transformation process view [41]. The development of these ontologies is an ongoing process, as the pace of scientific change, the rapid generation of new data, and the discovery of new concepts continuously drive the need for updates and expansions. The ontology serves as the backbone of structured knowledge, comprising four essential elements. First, classes represent overarching concepts or categories within a domain, such as “Voxel” in the design and engineering view (**Fig. 6a**) and “Transformation_Function”, “Transformation_Step”, and “Technical_Function” in the transformation process view (**Fig. 6b**). Second, individuals can be defined as specific instances of these classes, representing concrete entities (e.g., “1D_to_2D” as an instance of “Transformation_Dimension”, or “Repeatable” as an instance of “Transformation_Effect” from [41]). Third, object properties define relationships between individuals, establishing meaningful connections within the ontology (e.g., the relationship “isFulfilledBy” between the individuals of the classes “Transformation_Function” and “Transformation_Process”). Finally, data properties describe specific attributes or characteristics of classes (e.g., “Folding_angle” or “Bending_torque,” which are properties related to transformations like folding or bending) [41].

Given the increasing complexity and scale of data, there is a growing need for efficient methods to extend and refine existing ontologies. LLMs show significant potential in this regard. Trained on vast corpora, they are capable of generating, suggesting, and refining ontology terms based on their contextual understanding of language. This capability makes LLMs powerful tools for automating parts of the ontology expansion process, enabling more rapid adaptation to new scientific discoveries and evolving knowledge landscapes. By leveraging LLMs, ontology development can become more scalable and responsive, reducing the manual effort required while ensuring that ontologies remain comprehensive and up-to-date.

Alongside with ontologies, semantic networks represent knowledge as a graph, where nodes represent entities and edges represent relationships between them. These networks are used to capture the interconnectedness of concepts and facts. In 4D printing, a semantic network could model the relationships between different types of stimuli (e.g., temperature, light, moisture, etc.) and their effects on SMs (e.g., shape change, property change). For example, one node could represent an active polymer and another could represent the temperature. The edge between them would describe the reaction, e.g., the polymer expands when heated up to 60 °C. This interconnected representation allows for dynamic reasoning about how different stimuli will affect the printed object over time [149]. In addition, modern implementations of semantic networks in AI often take the form of knowledge graphs (KGs), leveraging technologies such as Neo4j [119], PuppyGraph [125], AWS Neptune [137], TigerGraph [160], OrientDB [121], ArangoDB [7], and NebulaGraph [118]. These tools provide powerful frameworks for representing and reasoning over complex relationships, which can be adapted to model the entire lifecycle of a 4D-printed object, including its design, printing, and behavior when exposed to stimuli [149,30].

3.2.3. Neuro-symbolic architectures

Neuro-symbolic systems have emerged by integrating the learning capacities of NNs with the structured reasoning strengths of symbolic AI. This integration enables the execution of logical inferences alongside data-driven learning. Building on the recent taxonomy proposed by Kautz [77] and later extended by Hamilton et al. [58] in terms of learning/reasoning capabilities, the existing NSAI architectures can be categorized into seven distinct types, as illustrated in **Fig. 7**. This figure synthesizes the relationships between neural and symbolic components through key integration strategies such as sequential, nested, ensemble, cooperative, and compiled. The classification includes the original six categories introduced by Kautz, which are expanded to include a seventh category (i.e., fibring), as proposed by Garcez and Gabby [50] and Hamilton et al. [58]. Each category represents a unique approach to integrating neural and symbolic methods, enabling the exploitation of their respective strengths:

- *symbolic Neuro symbolic* refers to systems where both input and output are represented in symbolic form, with a neural component acting as an intermediary processor. Symbolic knowledge, such as sequences of words or logical expressions, is first converted into vector representations, processed by a neural network, and then transformed back into a symbolic output. These architectures are particularly effective for tasks requiring symbolic reasoning enhanced by statistical generalization [84,41].
- *Symbolic[Neuro]* uses a neural component as a subroutine within a predominantly symbolic system. This allows statistical pattern recognition within a symbolic problem-solving framework. Such architectures are widely applied in robotics and autonomous systems, where symbolic planning integrates sensory data interpreted by neural networks [142].
- *Neuro[Symbolic]* embeds a symbolic reasoning engine within a neural system. This architecture is particularly effective for tasks that require a neural system to generalize from unstructured data while leveraging symbolic rules or relationships at certain conditions to enhance reasoning or interpretability. Therefore, it is capable of efficient combinatorial reasoning and currently represents one of the most promising approaches for combining the strengths of logic-based and neural-based AI [105,64].
- $Neuro|Symbolic$ processes non-symbolic input and converts it into a symbolic representation for reasoning, which in turn provides a feedback signal that is used to train the NN. For this case, the neural component – considered as a coroutine rather than a subroutine

Fig. 6. Extraction of a portion of an ontology on 4D printing [41]. (a) Design and engineering view with classes and object properties. (b) Transformation process view with classes and object properties. (Adapted with permission from Elsevier).

– can then learn from the errors of the symbolic component. Convergence is achieved through iterations. [107].

- *Neuro*$_{Symbolic}$ integrates symbolic rules as templates to guide the NN structure through weights or the loss function. The mapping of symbolic logic rules onto embeddings is compiled away during training [136,146,72].
- *Neuro:Symbolic* → *Neuro* incorporates symbolic rules during the training of NNs to guide their learning process. Using a structured training regime, neural models are trained to map input-output pairs without requiring step-by-step reasoning [131].
- *Fibring* operates on the principle of recursion, connecting multiple NNs through symbolic components [50].

Several notable algorithms have been developed in these categories. For instance, in the *Neuro*$_{Symbolic}$ category, logic tensor networks (LTNs) merge the power of first-order logic with NN learning. LTNs utilize tensors to represent logical predicates and relationships, allowing NNs to reason over data through logical inference and optimize through gradient descent. This architecture serves as a critical bridge between symbolic logic and neural learning, facilitating structured reasoning tasks [136]. Besides, tensor product representations (TPRs) encode symbolic entities and their relationships into distributed neural representations, allowing NNs to manipulate these structures within a tensor space. TPRs are critical for relational reasoning tasks, such as those involving multi-agent systems and knowledge representation [146]. In the *Neuro:Symbolic* → *Neuro* category, recursive neural knowledge networks (RNKNs)

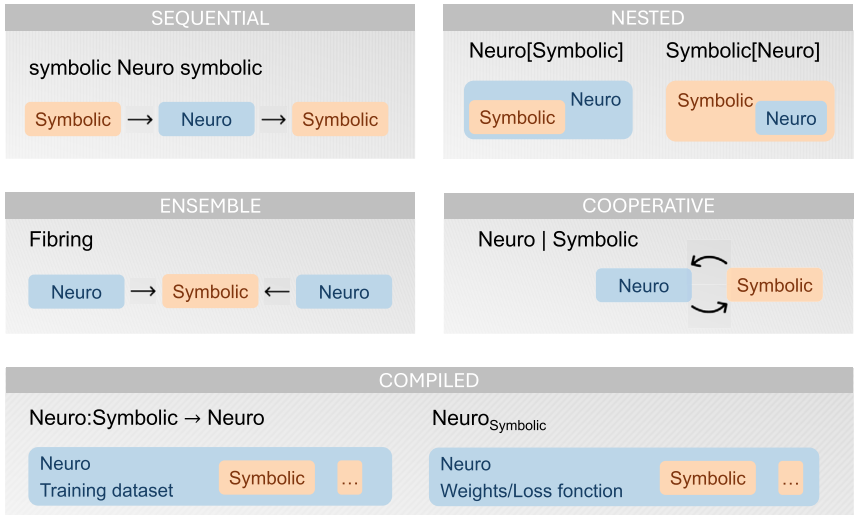

Fig. 7. Classification of NSAI architectures according to the relationships of the neural and symbolis components and their capabilities.

introduce symbolic knowledge into recursive NN structures. By recursively applying symbolic knowledge representations, RNKNs enable the propagation of knowledge across layers of reasoning, allowing for complex, multi-step logical inference. This is particularly beneficial for tasks that demand structured data reasoning [72]. In addition, logical neural networks (LNNs) directly incorporate symbolic reasoning within NN architectures. By embedding logical rules into the learning process, LNNs maintain logical consistency while still being trainable through gradient descent. This architecture is particularly well-suited for AI applications where adherence to predefined logical constraints is essential, such as in reasoning-based systems [131].

3.3. Explainability

NSAI addresses the challenges of traditional AI-based systems, by making AI models more transparent, interpretable, and trustworthy, without compromising their performance. Explainability provides insights into how inputs are processed, how decisions are made, and the reasoning behind those decisions. Symbolic reasoning contributes to this through logical and human-readable explanations of decisions. Typically in AI, the terms black box, gray box, and white box are utilized to describe the varying degrees of openness and transparency within a system. The black box problem in AI refers to the difficulty to justify and legitimize a decision. In contrast, a white box system ensures an internal decision-making process fully explainable.

Building on the foundation of explainable AI, NSAI integrates key explainability concepts that are crucial for addressing the black box problem and enhance trustworthiness in AI-based systems. Explainability, which allows human to comprehend how a system operates, is tightly linked to interpretability and extends into other principles. For instance, interactivity can foster a dynamic engagement between users and AI, enabling them to query and better understand its decisions. Transparency ensures that the decision-making process is clear and accessible, while responsibility emphasizes ethical considerations that ensure the system to align with societal values. Fairness guarantees that the AI system makes unbiased decisions across different population groups, and stability alongside satisfaction ensures the system's behavior remains consistent, with its explanations meeting user needs. Lastly, robustness refers to the system's ability to handle changes or uncertainties in its inputs without significantly altering its performance. All these interconnected principles work together to create a more reliable AI system, therefore fostering wider adoption for real-world applications.

3.4. Main achievements

NSAI has led to promising achievements across various fields. In NLP and automated theorem proving, NSAI transcends simple pattern recognition, enabling deep contextual understanding and precision in verifying complex proofs [139]. This approach has set new standards in solving intricate theorems that were previously out of reach. In healthcare, NSAI has accelerated drug discovery and diagnostics by bridging the gap between understanding complex biochemical interactions and predicting therapeutic outcomes. It offers interpretable insights, which not only streamline research but also improve treatment planning through personalized diagnostics. Similarly, in robotics and autonomous systems, NSAI empowers machines with both perceptual awareness and logical decision-making, enhancing their ability to navigate real-world environments safely and efficiently. Industries such as finance and law benefit from NSAI's capacity to automate structured information extraction from unstructured data, improving accuracy and speed in data-intensive processes. In supply chain management, NSAI enhances demand forecasting and resource allocation by integrating predictive insights with rule-based decision-making, optimizing operations in complex and variable conditions [150,157].

Some quantitative results can highlight NSAI's significant improvements in efficiency, scalability, and accuracy. Susskind et al. (2021) [156] evaluated the neuro-symbolic concept learner (NSCL) using the CLEVR dataset [75], a benchmark designed to assess visual reasoning and compositional generalization. The objective of NSCL is to combine visual understanding with reasoning capabilities by parsing scenes and questions into executable symbolic programs. They compared NSCL against purely neural models, specifically TbD-Net and MAC, which rely on end-to-end neural architectures for reasoning tasks. NSCL demonstrated a clear advantage, achieving 93% accuracy with only 10% of the training data, whereas TbD-Net and MAC plateaued at approximately 80% accuracy under the same conditions. This comparison highlights the superior data efficiency of NSAI, making it highly effective in scenarios with limited training data. In addition, Hersche et al. (2023) [62] introduced the neuro-vector-symbolic architecture (NVSA), whose objective is to combine NN representation learning with vector-symbolic algebra to address two key challenges: the "binding problem" in NNs, which limits their ability to disentangle and represent distinct objects within a compositional structure, and the computational inefficiency of exhaustive symbolic reasoning. NVSA leverages high-dimensional distributed representations to enable end-to-end learning that maintains both compositional transparency and computational efficiency. They applied NVSA to the I-RAVEN dataset [66], which focuses on solving

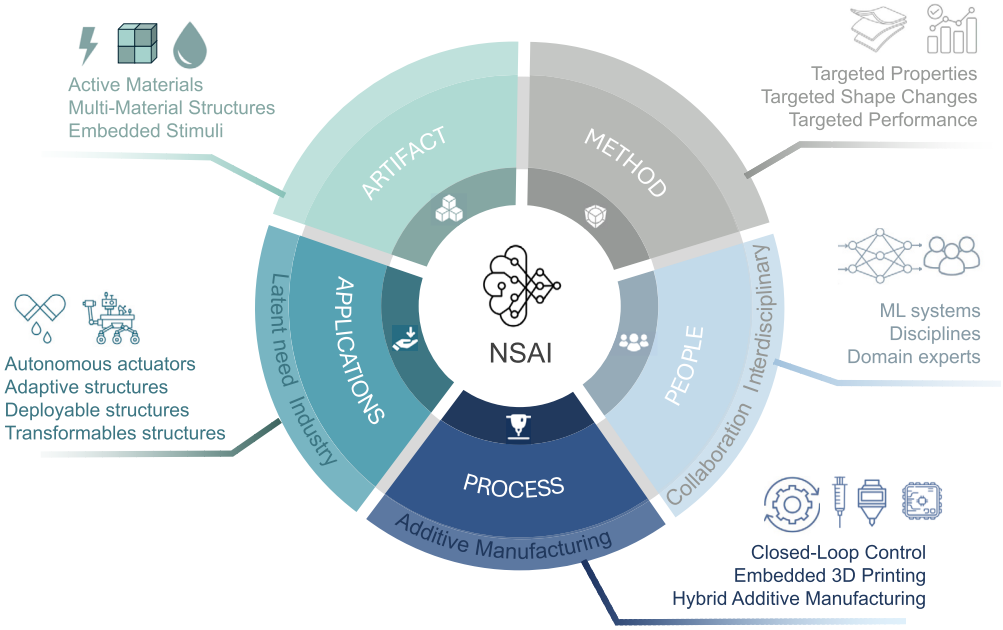

Fig. 8. Overview of NSAI in accelerated design for 4D printing with regard to AM, people involved from various fields, artifacts, and applications.

complex visual analogies, and compared it with baseline CNN models. NVSA achieved 88.1% accuracy while significantly reducing computational costs compared to these neural approaches. This highlights NVSA's effectiveness in tasks requiring both efficiency and accuracy.

Moreover, Han and Srivastava (2024) [60] focused on complex event detection (CED) using multimodal sensor data streams, such as inertial measurement unit (IMU) [174] and acoustic signals. The objective was to identify temporal patterns that signify high-level events, even when the data is noisy and spans long durations. They developed a neuro-symbolic event detection framework that integrates NNs for feature extraction with symbolic reasoning for temporal analysis. The framework was compared against widely-used neural methods, including LSTM, temporal convolutional models (TCM), and transformers, which are optimized for event detection tasks. Their neuro-symbolic approach demonstrated a 41% increase in F1 score, a widely-used metric in ML that balances precision (the proportion of correctly predicted positive events among all predicted positives) and recall (the proportion of correctly predicted positive events among all actual positives). This result highlights the framework's superior ability to combine contextual reasoning with data-driven insights, making NSAI particularly well-suited for applications requiring an understanding of temporal and causal relationships. Cunnington et al. (2024) [31] introduced a novel architecture called NeSyGPT, designed to integrate a vision-language foundation model (BLIP) with symbolic reasoning. The objective of NeSyGPT is to enable efficient neuro-symbolic learning by leveraging pre-trained vision-language models to extract symbolic features from raw data, followed by symbolic reasoning using answer set programming (ASP). This approach was applied across four problem domains: MNIST arithmetic [39], follow suit, plant hitting sets, and CLEVR-Hans, addressing tasks requiring both scalability and explainability. For example, in the follow suit domain, NeSyGPT demonstrated its ability to handle tasks with massive symbolic feature choices—scaling from 6.5 million in a 4-player card game to 5.74×10^{16} in a 10-player variant—while consistently achieving perfect accuracy. It significantly outperformed other neuro-symbolic methods, including NeurASP [179], SLASH [145], Embed2Sym [8], and BLIP+ABL [32], particularly in low-data scenarios. By integrating neural and symbolic components seamlessly, NeSyGPT demonstrated state-of-the-art scalability, accuracy, and explainability in highly demanding problem domains.

These findings underscore the major leap NSAI brings over conventional methods by uniting computational efficiency, advanced precision, and interpretative capabilities. NSAI not only addresses intricate real-world challenges more effectively, but also offers a transparent and adaptable framework—laying the groundwork for robust, next-generation AI systems. Given these achievements, it is essential to consider applying NSAI within 4D printing to harness its potential for managing the complexity of multi-material designs, stimuli-responsive structures, and real-time decision-making. NSAI could provide a transformative framework that could advance 4D printing applications across sectors that demand precision, adaptability, and reliability, setting the foundation for innovative and highly functional smart structures.

4. Opportunities in 4D printing design

The current achievements of NSAI and challenges in the design of active materials and transformable structures using 4D printing have highlighted the need for deeper exploration of material formulation and distribution, while maintaining control over AI-generated results. The following sections – illustrated in **Fig. 8** – present key opportunities in designing for 4D printing.

4.1. Multidisciplinary integration and coherence

The interdisciplinary nature of 4D printing presents substantial challenges in integrating expertise from diverse fields, such as engineering, design, materials science, AM, computational mechanics, design, chemistry, physics, and computer science. Each domain brings its own data, constraints, models, methodologies, and terminologies, making it difficult to establish cohesive and coordinated design processes. Demoly et al. (2021) [38] highlighted the importance of integrating multiple actors (e.g., product architects, mechanical designers, material experts, and AM process planners), while managing complexities such as multi-material structures, various AM techniques, functionalities, physical principles, scales, and temporal dimensions. NSAI is particularly valuable in addressing these challenges, as it facilitates collaborative design by enabling experts from different fields to work within a shared and explainable framework.

In industrial contexts, the multi-disciplinary integration is further complicated by distinct departments, such as engineering, manufacturing, and production, each with its specific goals, priorities, and methods. Unlike the discipline-oriented organization of research, industry departments are often project and task-driven and focused on addressing prac-

tical challenges. This distinction adds another layer of complexity to collaboration. NSAI bridges the gap between the theoretical focus of research and the practical needs of industry by providing a unified framework that fosters communication and collaboration across domains.

Achieving effective integration in 4D printing design requires both data interoperability and semantic alignment of knowledge from various sources. NSAI, with its dual capabilities of processing unstructured data (via NNs) and reasoning over structured rules (via symbolic reasoning), facilitates this alignment by creating a common ground for collaboration. For example, NSAI can integrate physics-based simulation results with material property datasets and user-defined design objectives into a coherent and dynamic decision-making framework. This capability not only enhances cross-disciplinary collaboration but also accelerates iterative design processes through more efficient exploration and refinement of design options. By supporting seamless communication, data integration, and informed decision-making, NSAI becomes a cornerstone for advancing collaborative and integrated design in 4D printing.

4.2. Fostering collaboration between ML and 4D printing experts

NSAI can play a transformative role in fostering collaboration ML systems and the diverse set of actors involved in 4D printing. A key challenge lies in bridging the opaque and data-driven nature of ML models with the clarity and precision required by experts to make informed decisions. Explainability ensures that the decision pathways of the ML model are accessible and interpretable to domain experts, allowing them to understand, validate, and modify how the model arrives at specific conclusions. This capability is crucial for establishing trust between 4D printing experts (i.e. designers, engineers, material expert, process planner, etc.) and AI systems, as it allows these experts to assess whether the model's decisions align with their domain knowledge.

In 4D printing, experts often possess domain-specific knowledge that is difficult to encode into traditional ML models. NSAI can address this by incorporating symbolic reasoning, allowing experts to embed their qualitative insights into the AI-driven decision-making process. This integration ensures that expert knowledge, which is based on years of experience and understanding of complex physical and/or chemical phenomena, can enhance and guide the ML models toward more accurate and contextually relevant predictions by explicitly incorporating expert-defined rules and constraints, which influence the resulting decisions of the model. These rules can help limit the model's outputs to feasible and realistic scenarios, ensuring that predictions align with practical constraints known to experts.

Explainability can also allow 4D printing experts to iteratively refine the AI model based on their insights. For instance, when an expert identifies a decision that does not meet practical expectations, they can trace back through the explainable components to understand why the decision was made, then update the symbolic reasoning layer accordingly. This iterative process helps the ML system evolve in a way that consistently incorporates new expert knowledge from new materials and structures development, making the model increasingly accurate and aligned with real-world needs.

4.3. Closed-loop additive manufacturing

ML techniques have attracted increasing attention in the field of AM due to their potential to optimize processes and improve printing quality. However, despite their promise, these approaches face several inherent limitations due to the complexity of AM techniques – especially when combined or hybridized, and the characteristics of the available data [124] [20]. These limitations can be grouped into five main aspects: process parameter variability, dependence on large datasets, inability to detect unknown defects, interpretability and integration challenges, and the high cost of real-time monitoring. AM processes, such as powder bed fusion (PBF), involve complex interactions between energy sources and materials, resulting in significant variability in parameters that affect part quality. With more than 130 parameters influencing printing, it is then difficult to accurately predict mechanical properties using ML techniques [90,63]. Traditional ML models show limited performance in the face of this complexity, often failing to capture all the dynamic interactions between these parameters. Next, training ML models, particularly those using deep NNs, requires large and diverse datasets to ensure accurate predictions. In the context of AM, collecting such data is challenging due to the diversity of materials, techniques, and geometries [113]. The data needed for modeling complex phenomena during AM can be costly to obtain, thus limiting the effectiveness of ML approaches.

Another significant limitation is the inability of supervised models to detect unknown defects. These models rely on the availability of historical labeled data and are therefore limited in their ability to identify anomalies or defects that have not been previously observed during training [120]. This represents a major constraint, as defects in AM, such as porosity, cracking, or delamination, can vary greatly depending on printing conditions. In addition, there are specific limitations associated with CNNs and image processing techniques in AM. CNNs have been widely used for defect detection and quality control by analyzing images from various stages of 3D printing process. However, CNNs require a large amount of labeled image data for training, which is often not available in AM [63]. CNNs can also be sensitive to changes in imaging conditions, such as lighting or camera angle, which can significantly impact the model's performance [63]. Image processing also tends to be computationally intensive, requiring substantial processing power and infrastructure to analyze high-resolution images in real-time. Additionally, CNNs often face difficulties in capturing subtle, small-scale defects, which may be critical in AM parts, potentially leading to undetected quality issues [113].

The lack of interpretability of ML models complicates the identification of defect causes and the implementation of corrective actions during the manufacturing process [89]. Integrating these models into existing AM systems presents challenges, particularly regarding the adaptation of equipment and operational practices. One can say that ML approaches applied to real-time monitoring of AM processes require sophisticated sensor systems and adapted infrastructure. These sensors must be capable of collecting in situ data with high resolution and precision, which increases the cost of implementing and maintaining these systems. Additionally, volumetric matching of validation data, such as X-ray computed tomography, remains difficult to achieve, thus limiting the effectiveness of ML models based on imaging [63].

These limitations illustrate the need to develop more robust ML approaches that can manage the variability and complexity of AM techniques, while reducing dependence on large datasets and improving model interpretability. In this context, NSAI can be employed to enhance the interpretability of ML models by integrating domain knowledge into the learning process, thereby making predictions more explainable. Furthermore, the symbolic reasoning component can help to identify and manage unknown defects by applying rule-based logic, even when sufficient labeled data is unavailable. NSAI also allows for more efficient use of available data, reducing the dependence on large datasets, and enabling more robust decision-making in real-time monitoring and optimization of AM processes. By using symbolic rules to fill in knowledge gaps, NSAI can operate effectively even with limited training data, which is particularly advantageous in 3D/4D printing scenarios where data collection is costly or time-consuming.

4.4. Optimizing multi-material and metamaterial structures

The development of mechanical metamaterials faces significant challenges in computational design and experimental validation, especially when coupled with ML technologies. The geometric complexity of these materials, characterized by hierarchical multi-scale structures such as lattices, plates, and shells, demands sophisticated computational models to predict their performance. Integrating complex architectural designs and printing parameters into these models is essential to meet

specific constraints, such as optimal stiffness, high thermal dissipation, lightweight, and low cost [82,18]. Numerical tools like finite element methods and topology optimization algorithms, including bidirectional evolutionary structural optimization (BESO) and the solid isotropic material with penalization (SIMP) method, enable progressive improvement of structures under complex constraints. However, these approaches are limited by vast design spaces and high computational demands [61,16,138].

ML offers a novel approach to material design, enabling the establishment of reporting standards and the use of benchmark datasets, its application in metamaterial optimization faces obstacles related to data quality, availability, and representativeness [116]. High-throughput simulations, such as those combining ML with molecular dynamics (MD), have shown promise in translating architected materials to the atomistic scale and discovering disordered mechanical metamaterials. However, these methods require extensive computational resources and rely on accurate forcefield models, which may not always generalize across different material systems [96]. Additionally, ML models often need large, diverse, and curated datasets to avoid biases and ensure reliable predictions. The lack of standardized datasets and unified frameworks for model evaluation further complicates the integration of ML into the metamaterial design process.

Finally, the experimental validation of ML-predicted metamaterials remains a bottleneck. Ensuring that the properties predicted by ML align with real-world performance requires precise fabrication techniques and rigorous testing methodologies. For example, advanced designs leveraging programmable Poisson's ratios and microstructural geometric adjustments [95] demand meticulous validation to confirm their in-situ tunability and mechanical robustness. These challenges underscore the importance of interdisciplinary strategies that combine advanced computational tools, innovative manufacturing processes, robust ML frameworks, and tailored experimental techniques.

Effective multi-scale topology optimization integrates both continuous and discrete optimization techniques, which is essential for addressing the complexities of multi-material design. NSAI not only reduces the solution space but also ensures that the optimization aligns with the specific requirements and constraints of the design, improving both functionality and printability. Additionally, NSAI supports bio-inspired strategies such as voxel automata, L-systems, and digital morphogenesis, which are vital for creating multi-scale structures [38]. By simulating growth patterns and evolutionary processes with NNs and applying structural and functional constraints through symbolic AI, this dual approach not only streamlines the optimization process for efficiency but also ensures it conforms to the specific requirements and constraints of multi-material designs, thereby enhancing functionality and printability.

NSAI can adopt an inverse design approach where NNs rapidly identify and adapt material distribution patterns based on targeted outputs such as target shape, mechanical strength, flexibility, and responsiveness to stimuli. NNs maximize mechanical and actuation performance by exploring large design spaces and generating configurations that leverage historical data to determine optimal distributions of active and passive materials within the 3D structure [154,153]. Symbolic AI then verifies and refines these patterns, ensuring alignment with physical laws and enforcing expert-driven design constraints. This hybrid capability supports voxel/octree-based modeling and uses rules based on expert knowledge to validate configurations, ensuring they meet design specifications and AM constraints. In cases of extreme environmental conditions, symbolic AI may recommend targeted adjustments to material placement, enhancing the structure's adaptation to fatigue and ensuring that multi-material structures remain robust, responsive, and reliable across diverse operational conditions.

In cases where datasets are limited, the integration of physics-informed NNs (PINNs) offers an effective solution. PINNs embed physical laws directly into NN models, enabling them to achieve higher precision in predicting material properties and interactions without relying on extensive data. However, while PINNs are excellent for incorporating physical laws, they can lack the versatility needed to handle complex design requirements that extend beyond pure physics. Symbolic AI becomes essential here, as it employs logical rules that represent expert-driven insights and practical constraints, guiding the optimization process in areas that PINNs alone may not sufficiently address. By combining these elements, NSAI validates predictions and embeds expert knowledge, offering a more adaptable and comprehensive approach that maintains high performance even when data is scarce or incomplete.

Integrating NSAI into the field of multi-material 4D printing provides a strategic pathway to overcoming current limitations. By harnessing the learning and adaptability of NNs alongside the logical, rule-based capabilities of symbolic AI, NSAI enables a holistic and scalable approach to designing, optimizing, and simulating multi-material structures across diverse scales. This approach not only accelerates discovery but also enhances the precision and practicality of designs in real-world applications. The future potential of NSAI in the realm of multi-material structures, metamaterials, and metastructures rests on its capacity to integrate predictive analytics with real-time adaptive capabilities. Advancing research should prioritize the incorporation of NSAI with real-time data acquisition and feedback mechanisms, facilitating continuous learning and structural adaptation over the lifecycle of a material. This progression could enable the creation of truly intelligent structures with self-monitoring and self-adjustment capabilities, unlocking unparalleled levels of performance, resilience, and longevity in complex and variable environments.

4.5. Exploring and discovery of new material compositions

The discovery and integration of new materials, particularly in multi-material and metamaterial designs, is one of the most promising areas in 4D printing. However, the vast design space of material composition presents a significant challenge. Understanding how different materials interact with each other and respond to stimuli requires careful exploration. While traditional NNs are highly effective at analyzing large datasets and identifying patterns in material behavior, they often lack the ability to provide transparent explanations or justify the use of specific material combinations. In 4D printing, knowing not just what material works but understanding why it works is essential for innovation. NSAI overcomes these limitations by integrating symbolic reasoning into the material discovery process. This allows the system not only to generate promising material combinations based on learned data but also to incorporate expert hypotheses derived from scientific literature and research. By combining NNs' ability to explore vast configurations of materials with symbolic AI's capacity to encode performance constraints such as cost, durability, and stimuli responsiveness, NSAI can systematically evaluate material combinations that satisfy multiple criteria.

For instance, experts' intuitive hypotheses, drawn from state-of-the-art research, can be encoded into the symbolic system to guide the exploration of material configurations. NSAI can then prioritize combinations based on criteria defined by symbolic reasoning, such as mechanical performance, environmental stability, or biocompatibility. As the NN proposes potential material candidates, the symbolic system evaluates these combinations to ensure they align with established scientific principles, thus bridging the gap between data-driven insights and knowledge-based reasoning.

Additionally, NSAI enhances the material discovery process by enabling the verification of predictions made by NNs through a structured symbolic reasoning framework. While NNs can generate potential material candidates by analyzing large datasets and identifying patterns, they may not provide insight into the chemical feasibility or performance of the predicted materials. This is where symbolic reasoning becomes crucial. The symbolic component of NSAI evaluates the proposed material combinations by applying knowledge-based rules derived from scientific principles, ensuring that the predicted materials are chemi-

cally viable and meet specific performance criteria. This knowledge-guided exploration helps verify whether the material will function as expected under real-world conditions, such as withstanding environmental changes or complying with printability constraints. Symbolic reasoning ensures that each material combination adheres to established scientific knowledge, including chemical properties, reactivity, and mechanical behavior, thus filtering out combinations that are unlikely to succeed. This process guarantees that the materials being considered for further development are not only innovative but also scientifically valid and technically feasible.

4.6. Discovery of emerging applications

In the realm of 4D printing, identifying practical applications remains a significant challenge. This difficulty arises from the disruptive and unconventional nature of printed shape-changing solutions, which often do not align seamlessly with existing market demands or user expectations. To bridge this gap, employing NSAI offers a promising avenue for discovering latent user needs, market experiences, and tacit desires that might not be immediately apparent to researchers involved in this emerging field [129]. Indeed, NSAI can explore both structured and unstructured data across industries, mining patent databases [141,135,140], industrial reports, ontologies and knowledge graphs, and scientific literature to uncover patterns, trends, and opportunities for new applications in 4D printing. Recent advances, including deep learning, AI-driven agent-based simulations, and LLM-based empathic mental inference, have demonstrated the potential to explore and analyze these sources for forecasting emerging economies [11], identifying and prioritizing customer needs, and eliciting design requirements [161,188,81,129]. By proactively identifying latent needs, technological gaps, and subsequenty emerging innovations, NSAI can position 4D printing as a transformative technology capable of addressing mid- and long-term real-world challenges. By capturing and understanding these intangible things, NSAI can guide 4D printing efforts towards discovering more promising and high-impact applications, ensuring that the technology addresses real-world challenges effectively.

Beyond industrial applications, NSAI can also explore other disciplines such as biology, physics, and chemistry to further develop the potential of 4D printing. For instance, by analyzing biological data and processes, NSAI can use NNs to identify hidden relationships and suggest biomimetic designs, and uncover biological principles that can be replicated with smart materials and stimuli. This cross-disciplinary capability allows for the creation of solutions that go beyond traditional cases developed in the literature, therefore applying 4D printing in areas like tissue engineering, smart implants, and responsive drug delivery systems, among others.

5. Conclusion

The integration of NSAI into 4D printing represents a critical opportunity to address the complexities of multi-material design, stimuli responsiveness, and interdisciplinary collaboration. As 4D printing is a booming and interdisciplinary research field, the need for precise material behavior control, scalable design frameworks, and transparent decision-making processes becomes increasingly important. NSAI not only bridges gaps between fields such as materials science, chemistry, AM, mechanics, engineering, and design optimization but also enhances the capabilities of existing computational methods, ensuring that design decisions are grounded in logical reasoning and backed by comprehensive datasets. By synthesizing diverse data from various domains and offering explainable recommendations, NSAI allows involved experts to navigate the intricate interactions between materials and stimuli with a level of precision that traditional approaches cannot achieve. This transparency in decision-making fosters greater collaboration across disciplines, optimizing both material composition, distribution, printing, and stimuli control at multiple scales. Ultimately, NSAI unlocks new possibilities for innovation, enabling the development of 4D-printed structures that are more reliable, efficient, and tailored to meet the specific demands of diverse industries, from biomedical engineering to automotive and space. As the field moves forward, the adaptability and comprehensive decision-making capabilities of NSAI will play a pivotal role in driving the future of 4D printing.

CRediT authorship contribution statement

Oualid Bougzime: Writing – original draft, Methodology, Investigation. **Christophe Cruz:** Writing – review & editing, Supervision, Methodology. **Jean-Claude André:** Writing – review & editing, Methodology. **Kun Zhou:** Writing – review & editing, Methodology. **H. Jerry Qi:** Writing – review & editing, Methodology. **Frédéric Demoly:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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